

Project Title: Balaji Fast Food Sales

Team Name: Team Ignis

Team Details:

1. L. Pranay(Team lead) - 23341A1266
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4. V. Bhargav Raju - 23341A12C9

Repository Links:

- 1.<https://github.com/pranaylekkala5390/GenAI-Assignment>
- 2.<https://github.com/srikar0119/GenAI-Assignment>
- 3.<https://github.com/Haneesh-Srirama/GenAI-Assignments>
- 4.<https://github.com/bhargav123321/GenAI-Assignments>

Assignments:Total(10)

1. L. Pranay - 07
2. P. Srikar - 07
3. S. Haneesh - 06
4. V. Bhargav Raju - 07

Project Updates:

1.Data Inspection:

Dataset Overview:

Number of rows: 1,000
Number of columns: 10
Dataset dimensions: 1,000 × 10
Total cells: 10,000
Dataset size: 10,000 values/cells
Columns: 10
Duplicate rows: 0

Column names:

1. order_id
2. date
3. item_name
4. item_type
5. item_price
6. quantity
7. transaction_amount
8. transaction_type
9. received_by
10. Time_of_sale

Data Structure:

Data types:

order_id – int64

date – object

item_name – object

item_type – object

item_price – int64
quantity – int64
transaction_amount – int64
transaction_type – object
received_by – object
time_of_sale – object

Non-null observations:

order_id – 1,000
date – 1,000
item_name – 1,000
item_type – 1,000
item_price – 1,000
quantity – 1,000
transaction_amount – 1,000
transaction_type – 893
received_by – 1,000
time_of_sale – 1,000

Categorical Summary

1.date:

Start Date: 01-04-2022
End Date: 30-03-2023
Number of unique values: 348
Most frequent date: 05-09-2022
Frequency: 8

2.item_name:

Number of unique values: 7
Most frequent item: Cold coffee
Frequency: 161

3.item_type:

Number of unique values: 2
Most frequent category: Fastfood
Frequency: 686

4.transaction_type;

Number of recorded categories: 2
Most frequent transaction type: Cash
Frequency: 476

5.received_by:

Number of unique values: 2
Most frequent category: Mr.
Frequency: 512

6.time_of_sale:

Number of unique values: 5
Most frequent time period: Night
Frequency: 205

2.Data Preprocessing:

Handling Missing Values:

Missing values were checked using `df.isnull().sum()`.

Observation: No missing values were present after preprocessing.

Duplicate Records:

Duplicate records were checked using `df.duplicated().sum()`.

Observation: 0 duplicate records were found.

Date Conversion:

The `date` column was converted from object type to datetime format.

Necessity: To perform date-based and time-series analysis.

Negative Value Check:

Negative values were checked in:

- `item_price`
- `quantity`
- `transaction_amount`

Observation: No negative values were found.

Zero Value Check:

Zero values were checked in:

- `item_price`
- `quantity`
- `transaction_amount`

Observation: No zero values were found.

Transaction Amount Validation:

A temporary `calculated_amount` column was created using:

$\text{item_price} \times \text{quantity}$

The calculated values were compared with `transaction_amount`.

Observation: No mismatches were found.

Categorical Value Check:

The categorical columns were checked for unique and valid values.

Observation: The dataset contained valid categories for item name, item type, transaction type, received by, and time of sale.

Outlier Detection:

Outliers were detected in `transaction_amount` using the IQR method.

Lower Bound: -240

Upper Bound: 720

Number of Outliers: 44

The identified outliers were retained because they represent valid high-value transactions.

Final Dataset:

After preprocessing:

Rows: 1,000

Columns: 10

Missing Values: 0

Duplicate Records: 0

Date Type: datetime64[ns]

3.Data Visualization:

Product-wise Sales:

Chart: Bar Chart

Purpose: To compare the total sales generated by each product.

Observation: The chart shows the sales performance of individual products and helps identify high- and low-performing products.

Revenue by Item Type:

Chart: Bar Chart

Purpose: To compare revenue generated by different item categories.

Observation: The chart shows which item type contributes more to total revenue.

Quantity Sold by Product:

Chart: Bar Chart

Purpose: To compare the quantity of each product sold.

Observation: The chart helps identify the most and least frequently sold products.

Transaction Type Distribution:

Chart: Pie Chart

Purpose: To understand the distribution of different transaction types.

Observation: The chart shows the proportion of Cash, Online, and Unknown transactions.

Sales by Time of Sale:

Chart: Bar Chart

Purpose: To identify the time period generating the highest sales.

Observation: The chart helps identify the busiest sales periods.

Monthly Sales Trend:

Chart: Line Chart

Purpose: To analyze changes in sales over different months.

Observation: The line chart shows the overall sales trend and variations across months.

Transaction Amount Distribution:

Chart: Histogram

Purpose: To understand the distribution and frequency of transaction amounts.

Observation: The histogram shows how transaction amounts are distributed across different ranges.

Transaction Amount Outliers:

Chart: Box Plot

Purpose: To visually identify unusually high or low transaction amounts.

Observation: The box plot highlights the outliers identified using the IQR method.

Sales by Received By:

Chart: Bar Chart

Purpose: To compare sales across the `received_by` categories.

Observation: The chart shows the difference in sales associated with each category.